

Effect of Hydrogen Bonding between Ions of Like Charge on the Boundary Layer Friction of Hydroxy-Functionalized Ionic Liquids

Hua Li, Thomas Niemann, Ralf Ludwig, and Rob Atkin

J. Phys. Chem. Lett., **Just Accepted Manuscript** • DOI: 10.1021/acs.jpclett.0c00689 • Publication Date (Web): 27 Apr 2020

Downloaded from pubs.acs.org on April 30, 2020

Just Accepted

“Just Accepted” manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides “Just Accepted” as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. “Just Accepted” manuscripts appear in full in PDF format accompanied by an HTML abstract. “Just Accepted” manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). “Just Accepted” is an optional service offered to authors. Therefore, the “Just Accepted” Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the “Just Accepted” Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these “Just Accepted” manuscripts.

Effect of Hydrogen Bonding between Ions of Like Charge on the Boundary Layer Friction of Hydroxy-Functionalized Ionic Liquids

Hua Li,^{1,2,#,*} Thomas Niemann,^{3,4#} Ralf Ludwig,^{3,4,5} Rob Atkin,¹

¹*School of Molecular Sciences, The University of Western Australia, Perth, Western Australia 6009, Australia*

²*Centre for Microscopy, Characterisation and Analysis, The University of Western Australia, Perth, Western Australia 6009, Australia*

³*Universität Rostock, Institut für Chemie, Abteilung für Physikalische Chemie, Dr.-Lorenz-Weg 2, 18059, Rostock, Germany*

⁴*Department LL&M, University of Rostock, Albert-Einstein-Str. 25, 18059, Rostock, Germany*

⁵*Leibniz-Institut für Katalyse an der Universität Rostock e.V., Albert-Einstein-Str. 29a, 18059 Rostock, Germany*

[#]These authors contribute equally.

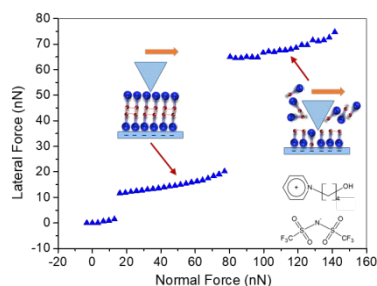
E-mail: hua.li@uwa.edu.au

Abstract

Atomic force microscopy (AFM) has been used to measure the lubricity of a series of ionic liquids (ILs) at mica surfaces in the boundary friction regime. A previously unreported cation bilayer structure is detected at the IL-mica interface due to the formation of H-bonds between the hydroxy-functionalized cations ((c-c) H-bonds), which enhances the ordering of the ions in the boundary layer and improves the lubrication. The strength of the cation bilayer structure is controlled by altering the strength of (c-c) H-bonding via changes in the

hydroxyalkyl chain length, cation charge polarizability and the coordination strength of the anions. This reveals a new means of controlling IL boundary nanostructure via H-bonding between ions of the same charge, which can impact diverse applications including surface catalysis, particle stability and electrochemistry, etc.

TOC Graphic



1
2
3 Ionic liquids (ILs) are pure salts with low melting points.¹⁻⁵ IL properties are controlled by a
4 subtle balance of Coulombic, dispersion, and H-bonding forces,⁶⁻¹⁰ which can be tuned by
5 altering the cation or anion charge group type and the length or number of attached organic
6 chains.^{4, 11} Due to their designable nature, ILs have widespread applications,^{12 13-18} many of
7 which involve processes at interfaces, so are at least partially controlled by interfacial ion
8 arrangements.

9
10
11
12
13
14
15
16
17
18
19 IL ions interact strongly with solid surfaces, and often have pronounced interfacial structure,
20 which can be divided into three distinct regions: the boundary or Stern (surface-adsorbed)
21 layer, the transition zone (near-surface layers) and the bulk phase.¹⁹⁻²⁴ The boundary layer
22 consists of ions in direct contact with the solid surface. The composition of the boundary
23 layer strongly depends on the properties of the solid substrate. Adjacent to the boundary layer
24 are several near-surface layers, referred to as the transition zone. Through this transition zone,
25 which is typically a few nanometres across, the pronounced interfacial layer structure decays
26 into the bulk morphology.

27
28
29
30
31
32
33
34
35
36
37
38
39
40 Atomic force microscopy (AFM) has been used to investigate a variety of IL-solid interfaces.
41 In a typical AFM normal force measurement, as the AFM probe approaches from the bulk IL
42 to the solid interface, a series of pronounced discrete steps in the force data are due to
43 rupturing near-surface layers.¹⁷ The ‘push-through force’ for an IL layer is the force
44 maximum of each step; a larger push-through force indicates a higher degree of ordering (that
45 is, stronger cohesive interactions) in the ion layer. Since AFM can measure the normal force
46 and lateral (frictional) force simultaneously (via tracking the normal and lateral deflections of
47 the AFM cantilever), it is also an excellent technique to investigate friction at the
48 nanoscale.²⁵⁻²⁶ When the adhesion between the two contacting solid surfaces is not significant,
49
50
51
52
53
54
55
56
57
58
59
60

lateral (frictional) force generally increases linearly with normal force. The proportionality between the lateral force and the normal force is given by the friction coefficient (μ), which is a useful parameter in comparing the lubrication properties between systems.²⁷

H-bonding in ILs is typically between a cation and an anion (c-a). However, H-bonding between ions of like charge, has recently been detected in the bulk phase of ILs with hydroxy-functionalized imidazolium and pyridinium cations, leading to the formation of H-bonded cation (c-c) clusters.²⁸⁻³² In the bulk, (c-c) H-bonding can be tuned via the coordination strength of the anion, the polarizability (charge delocalization) of the cation, and the length of the hydroxyalkyl chain.^{7, 33} For example, 1-(4-hydroxybutyl)pyridinium bis(trifluoromethanesulfonyl)imide ($[\text{HOC}_4\text{Py}][\text{NTf}_2]$) forms strong (c-c) H-bonds. Replacing the weakly coordinating $[\text{NTf}_2]^-$ anion with the strongly coordinating methanesulfonate ($[\text{OMs}]^-$) anion leads exclusively to (c-a) H-bonding, although the cation is the same. Replacing the delocalised charge on the polarisable pyridinium cation of $[\text{HOC}_x\text{Py}][\text{NTf}_2]$ with the charge-localised piperidinium cation ($[\text{HOC}_x\text{MPip}]$), results in less effective compensation of the anion charge centre and weaker (c-c) H-bonds. Tethering the OH group close to the positive charge centre of the cationic ring with shorter hydroxyalkyl chains enhances the like-charge repulsion thus reduces the strength of (c-c) H-bonding.

Previous AFM imaging studies have shown that the bulk IL nanostructure is flattened by the presence of a smooth solid interface,³⁴ analogous surface induced sponge to lamellar phase transitions for aqueous surfactants.³⁵ Similar behaviours have been reported for a set of hydroxy-functionalized ILs at a mica surface, where (c-c) H-bonds enhance the interfacial structure of ILs and stabilize the near-surface layers.³⁶ AFM normal force curves revealed that $[\text{HOC}_4\text{Py}][\text{NTf}_2]$ has the strongest structure of the ILs investigated owing to the strong

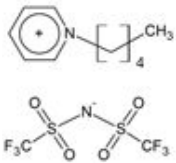
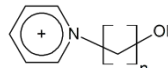
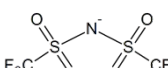
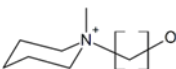
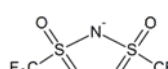
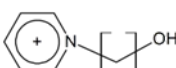
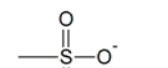
1
2
3 stabilizing effect of (c-c) H-bonding between the hydroxy-functionalized $[\text{HOC}_4\text{Py}]^+$ cations,
4
5 as well as weak coordinations between $[\text{HOC}_4\text{Py}]^+$ cations and $[\text{NTf}_2]^-$ anions.
6
7 $[\text{HOC}_4\text{MPip}][\text{NTf}_2]$ has weaker interfacial structure, as the localized positive charge on the
8
9 $[\text{HOC}_4\text{MPip}]^+$ cation enhances Coulomb repulsion and thus destabilizes the (c-c) H-bonds.
10
11 $[\text{HOC}_x\text{Py}][\text{OMs}]$ shows the weakest interfacial structure as the strongly coordinating $[\text{OMs}]^-$
12
13 anions form almost entirely anion-cation pairs.
14
15
16
17
18

19 Mica surfaces have a high density of permanent negative charge sites on its surface. Previous
20
21 AFM studies have shown that even at the highest force an AFM probe with a given spring
22
23 constant can impart (contact stress in the range of GPa) at least one strongly adsorbed cation
24
25 layer exists at the interface and resists “squeeze out”.^{23, 34, 37-38} At compliance, the AFM
26
27 probe pushes against an adsorbed cation layer rather than the solid surface. Based on the
28
29 similarity of the cation charge groups, it is likely this also occurs for hydroxy-functionalized
30
31 cations. That is, after pushing through the last measurable near surface ion layer, the AFM
32
33 probe touches on a strongly adsorbed boundary layer rich in cations rather than mica. The
34
35 consequence of this is that AFM force curves provide no direct information on the properties
36
37 of the boundary layer. However, as boundary layer friction is controlled by species adsorbed
38
39 to a surface, AFM nanotribology studies can directly probe ions that cannot be displaced by
40
41 the AFM probe at high force.²⁵⁻²⁶
42
43
44
45
46
47
48

49 In this work, AFM is used to measure the boundary layer friction at mica surfaces for the six
50
51 hydroxy-functionalized ILs, and one unfunctionalized IL, shown in Table 1. The results are
52
53 discussed in comparison with the infrared spectroscopy (IR) data (SI Figure 1-3)^{7, 31, 39-40}
54
55 and AFM normal force curves³⁶. The outcomes will pave the way for controlling IL boundary
56
57
58
59
60

nanostructure via tuning the strength of H-bonds between like-charged ions, which will underpin the development of new advanced lubricants.

Table 1. Full name, abbreviation, and structure of the unfunctionalized and hydroxy-functionalized ILs investigated in this study.

IL Full Name	IL Abbreviation	IL Structure
1-pentylpyridinium-bis(trifluoromethanesulfonyl)imide	[C ₅ Py] [NTf ₂]	
1-(3-hydroxypropyl)pyridinium-bis(trifluoromethanesulfonyl)imide	[HOC ₃ Py] [NTf ₂]	
1-(4-hydroxybutyl)pyridinium-bis(trifluoromethanesulfonyl)imide	[HOC ₄ Py] [NTf ₂]	
1-(3-hydroxypropyl)methylpiperidinium-bis(trifluoromethanesulfonyl)imide	[HOC ₃ MPip] [NTf ₂]	
1-(4-hydroxybutyl)methylpiperidinium-bis(trifluoromethanesulfonyl)imide	[HOC ₄ MPip] [NTf ₂]	
1-(3-hydroxypropyl)pyridinium-methanesulfonate	[HOC ₃ Py] [OMs]	
1-(4-hydroxybutyl)pyridinium-methanesulfonate	[HOC ₄ Py] [OMs]	

The ILs used in this work were synthesized at University of Rostock, Germany. The synthesis methods were described in detail elsewhere.³⁶ The ILs were dried under vacuum to a water content below 10 ppm before use. Lateral (frictional) force measurements were performed using a Veeco Nanoscope IV AFM in contact mode with an EV scanner. Sharp silicon AFM probe (spring constant = 0.8 ± 0.2 N/m) from the same batch (model NSC36, Mikromasch, Tallinn, Estonia) were used over the course of the investigation. The ILs were held in an

AFM fluid cell, sealed by a silicone O-ring. Lateral forces were obtained by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled.⁴¹ The lateral deflection signal (i.e., cantilever twist) was converted to lateral force using a customized function produced in Matlab.^{26, 42} The experiments were repeated more than three times; the obtained lateral force data showed the same trend with acceptable variations.

Figure 1(a) shows lateral force (F_L) versus normal force (F_N) of the unfunctionalized IL $[C_5Py][NTf_2]$ at a mica surface. Like previous studies with similar ILs,⁴³ two friction regimes are apparent: the low normal force multilayer regime, where multiple ion/ion pair layers separate the AFM probe and the surface; and the boundary regime, where the AFM probe slides against the boundary layer. The critical normal force (breakpoint) which delineates the transition from the multilayer regime to the boundary regime detected in Figure 1(a) is 3 nN; the contact stress (p_{max}) calculated using the equation in the SI is 0.02 GPa. This normal force is in good agreement with the final push-through force of 2 nN in the normal force curve (Figure 1(a) inset).³⁶ In the multilayer regime, friction is affected by many factors, including adhesion, the structure of near surface layers, etc, which are not the focus of this study. The following discussion is primarily concerned with the friction in the boundary layer regime, when the AFM probe is in contact with a strongly adsorbed ion layer. In the boundary regime of $[C_5Py][NTf_2]$ ($F_N > 3$ nN), the lateral force increases linearly with the normal force due to reduced adhesion. The friction coefficient of $[C_5Py][NTf_2]$ in the boundary regime has a relatively small value of 0.25 (Table 2). This means the boundary layer enriched in $[C_5Py]^+$ cations is smooth, resulting in low friction.

The lubricity of the hydroxy-functionalized ILs is completely different from (the unfunctionalized) $[\text{C}_5\text{Py}][\text{NTf}_2]$. For $[\text{HOC}_4\text{Py}][\text{NTf}_2]$, three regimes with two critical normal forces (breakpoints) at 12 nN ($p_{\text{max}}=0.09$ GPa) and 81 nN ($p_{\text{max}}=0.60$ GPa), respectively are detected, as seen in Figure 1(b). The first breakpoint (12 nN) roughly corresponds to the final push-through force in the AFM normal force curves (Figure 1(b) inset).³⁶ This means when the normal force is higher than 12 nN, the AFM probe should have reached the boundary regime, where a strongly adsorbed cation boundary layer resists “squeeze out” and the lateral force increases monotonously with normal force. However, for $[\text{HOC}_4\text{Py}][\text{NTf}_2]$ a second breakpoint is detected at a normal force of 81 nN, indicating rupture of an additional layer. This raises one question as to whether the strongly adsorbed cation-enriched boundary layer has squeezed out resulting in direct contact of the AFM probe and mica at high normal force. Normal force curves suggest this is unlikely: the near surface layers of $[\text{HOC}_4\text{Py}][\text{NTf}_2]$ are more difficult to rupture than $[\text{C}_5\text{Py}][\text{NTf}_2]$, due to strong stabilising H-bonding between the $[\text{HOC}_4\text{Py}]^+$, as detected in bulk IR spectra (SI Figure S1(a)) and AFM normal force curves³⁶. It is therefore expected that, in the boundary layer, $[\text{HOC}_4\text{Py}]^+$ cations should be more difficult to squeeze out than $[\text{C}_5\text{Py}]^+$ cations due to stronger (c-c) H-bonding. Therefore, for $[\text{HOC}_4\text{Py}][\text{NTf}_2]$, when the AFM normal force curves reach compliance it is probable that the AFM probe pushes against a bilayer structure enriched in $[\text{HOC}_4\text{Py}]^+$ cations due to the formation of cation clusters via (c-c) H-bonding, as illustrated in Scheme 1(a). It should be noted that this schematic provides only an indication of the most probable ion arrangements. In fact, the ions are in constant motion, and diffuse in and out of the layer, so other ion arrangements undoubtedly occur, but are less probable. During the friction measurement, the first $[\text{HOC}_4\text{Py}]^+$ cation layer could be shaved away when F_N is higher than 81 nN owing to the destruction of (c-c) H-bonds between the two cation layers, leaving the final $[\text{HOC}_4\text{Py}]^+$ cation layer confined between the AFM probe and mica which prevents direct surface contact

(Scheme 1(b)). The sudden jump in lateral force at the breakpoints are proportional to the energy dissipated to shave away the ion layers adsorbed on top of the sliding plane.⁴⁴ For [HOC₄Py][NTf₂], the sudden jump in lateral force at the 2nd breakpoint (~ 40 nN) is much higher than at the 1st breakpoint (~ 10 nN), indicating that (c-c) H-bonds in the boundary bilayers are difficult to break and more energy is required to shave away the first [HOC₄Py]⁺ layer of the boundary bilayer than the final near surface layer.

The friction coefficient of [HOC₄Py][NTf₂] in the bilayer regime (between normal forces of 12 nN and 81 nN) is 0.12, which is more than 50% lower than for [C₅Py][NTf₂] in the boundary regime (0.25). This indicates the [HOC₄Py]⁺ bilayer presents a more ordered, smoother sliding plane to the AFM probe than a [C₅Py]⁺ enriched monolayer due to directional (c-c) H-bonding. The friction coefficient of [HOC₄Py][NTf₂] increases slightly above 81 nN when only one cation enriched layer separates the AFM probe and mica. This could be due to the [HOC₄Py]⁺ monolayer being less ordered than the bilayer, or H-bonds between the silanol groups of the AFM probe and the hydroxyl groups of the cations.

The lubricity of [HOC₃Py][NTf₂] is similar to [HOC₄Py][NTf₂]. Three regimes are also detected: the low normal force (multilayer) regime, the medium normal force (cation bilayer) regime and the high normal force (cation monolayer) regime, respectively. The only differences are both the critical normal forces (breakpoints) and the sudden jumps in the lateral forces are lower for the shorter alkyl chain. This indicates that both the multilayers and the cation bilayers are less stabilised and more easily shaved away for [HOC₃Py][NTf₂] than [HOC₄Py][NTf₂]. This is primarily due to weaker (c-c) H-bonding caused by the stronger like-charge repulsions for shorter alkyl chains, similar with the effects found in the bulk (SI Figure S1(b))^{7, 33} and at the near surface (Figure 1 inset)³⁶. The friction coefficients in both cation

bilayer regime and monolayer regime are $\sim 20\%$ higher for $[\text{HOC}_3\text{Py}][\text{NTf}_2]$ than for $[\text{HOC}_4\text{Py}][\text{NTf}_2]$, as shown in Table 2. This suggests that the reduced (c-c) H-bonding reduces the ordering of cations in both the bilayer and the monolayer.

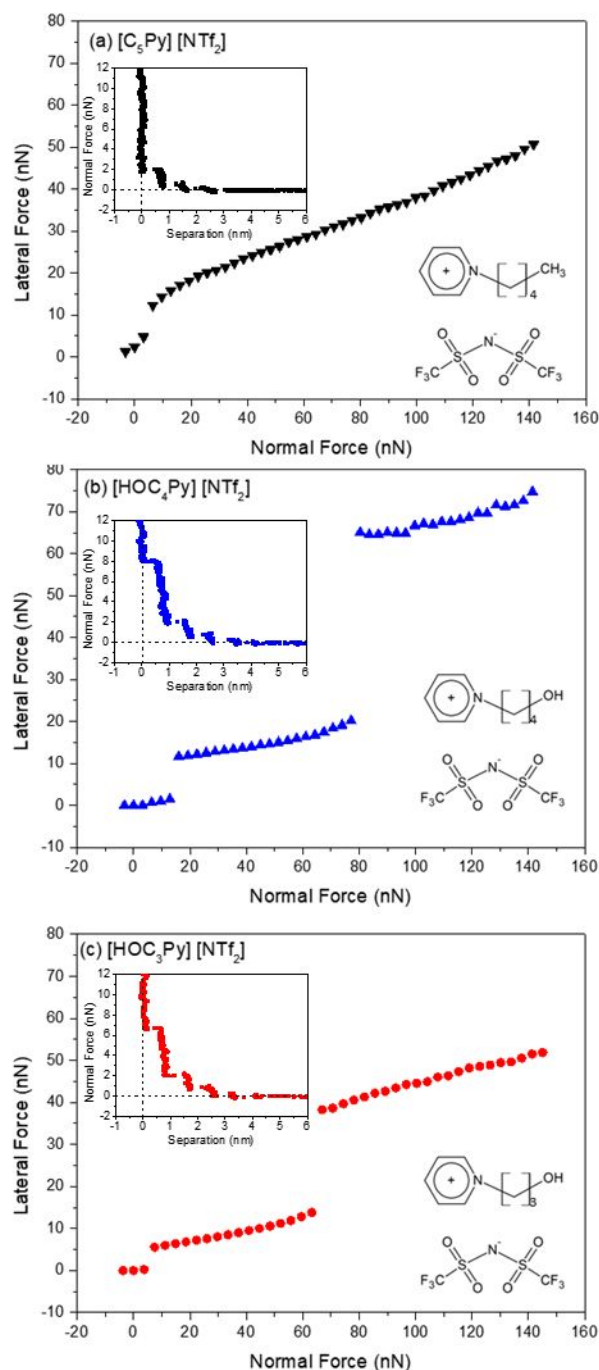
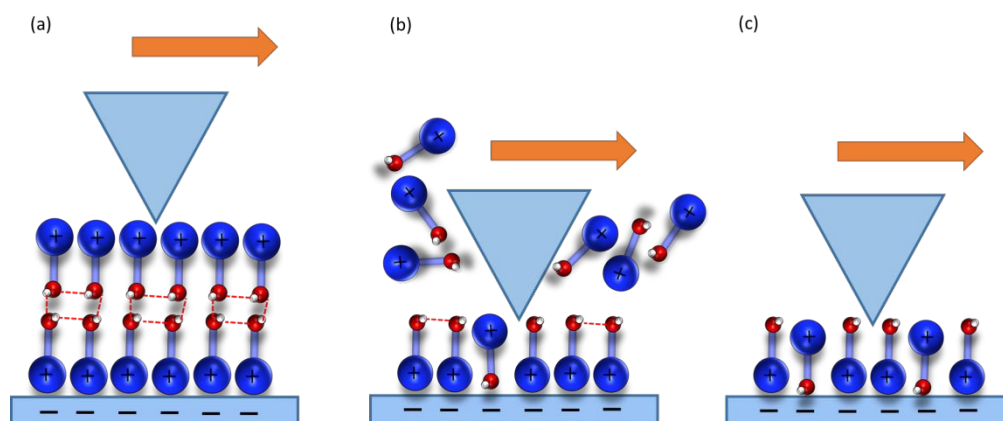


Figure 1. Lateral force vs normal force of (a) $[\text{C}_5\text{Py}][\text{NTf}_2]$, (b) $[\text{HOC}_4\text{Py}][\text{NTf}_2]$ and (c) $[\text{HOC}_3\text{Py}][\text{NTf}_2]$ at mica surfaces. Inset: normal force curve of the same IL³⁶.



Scheme 1. (a) an AFM probe slides on a cation bilayer formed at the negatively charged mica surface due to (c-c) H-bonding; (b) an AFM probe slides on a single cation layer when the top of the cation bilayer is ruptured, and (c) an AFM probe slides on a single cation layer without (c-c) H-bonding.

Figure 2(a) shows the effect of reducing cation charge delocalization (polarizability) on lateral forces using $[\text{HOC}_x\text{MPip}][\text{NTf}_2]$ ILs. For both $[\text{HOC}_4\text{MPip}][\text{NTf}_2]$ and $[\text{HOC}_3\text{MPip}][\text{NTf}_2]$, the first breakpoints occur at very low normal forces ($F_N < 3$ nN), which are consistent with the final push through forces detected in AFM normal force curves.³⁶ This shows reducing cation charge delocalization weakens the (c-c) H-bonding and thus layer cohesion near the surface. The second breakpoints occur also at lower normal forces, and the jump in force at the breakpoint is less than for $[\text{HOC}_x\text{Py}][\text{NTf}_2]$ ILs. This means the integrity of the cation bilayer structure in the boundary regime also decreases when charge localisation is increased, as this weakens (c-c) H-bonding. The friction coefficients in both medium and high normal force regimes are also higher for $[\text{HOC}_x\text{MPip}][\text{NTf}_2]$ than $[\text{HOC}_x\text{Py}][\text{NTf}_2]$ of the same chain lengths, indicating less ordered cation bilayers and monolayers due to weakened (c-c) H-bonding.

The lubricity of $[\text{HOC}_x\text{Py}][\text{OMs}]$ ILs is similar to the unfunctionalized IL $[\text{C}_5\text{Py}][\text{NTf}_2]$ (Figure 2(b)). The $[\text{OMs}]^-$ anion strongly coordinates to the OH functional group of $[\text{HOC}_x\text{Py}]^+$, resulting in strong, Coulomb enhanced (c-a) H-bonds, and the almost complete absence of stabilizing (c-c) H-bonds, as shown in the IR spectra (SI Figure S1(c)).

Consequently, no cation clusters or bilayers exist near the surface for $[\text{HOC}_x\text{Py}][\text{OMs}]$ and the lubricity is similar to conventional ILs.^{7, 36} This indicates the extra OH functional group on the cation alkyl chain has no obvious effects in cation orientation and ordering in the boundary regime, most likely due to the elimination of (c-c) H-bonding in the boundary layers, as shown in Scheme 1(c).

Table 2. Friction coefficients (μ) of $[\text{C}_5\text{Py}][\text{NTf}_2]$ and $[\text{HOC}_n\text{Py}][\text{NTf}_2]$ ILs.

	$[\text{C}_5\text{Py}]$ $[\text{NTf}_2]$	$[\text{HOC}_4\text{Py}][\text{NTf}_2]$		$[\text{HOC}_3\text{Py}][\text{NTf}_2]$		$[\text{HOC}_4\text{MPip}][\text{NTf}_2]$		$[\text{HOC}_3\text{MPip}][\text{NTf}_2]$		$[\text{HOC}_4\text{Py}]$ [OMs]	$[\text{HOC}_3\text{Py}]$ [OMs]
		Medium force	High force	Medium force	High force	Medium force	High force	Medium force	High force		
μ	0.25	0.12	0.15	0.14	0.18	0.16	0.17	0.23	0.21	0.24	0.26

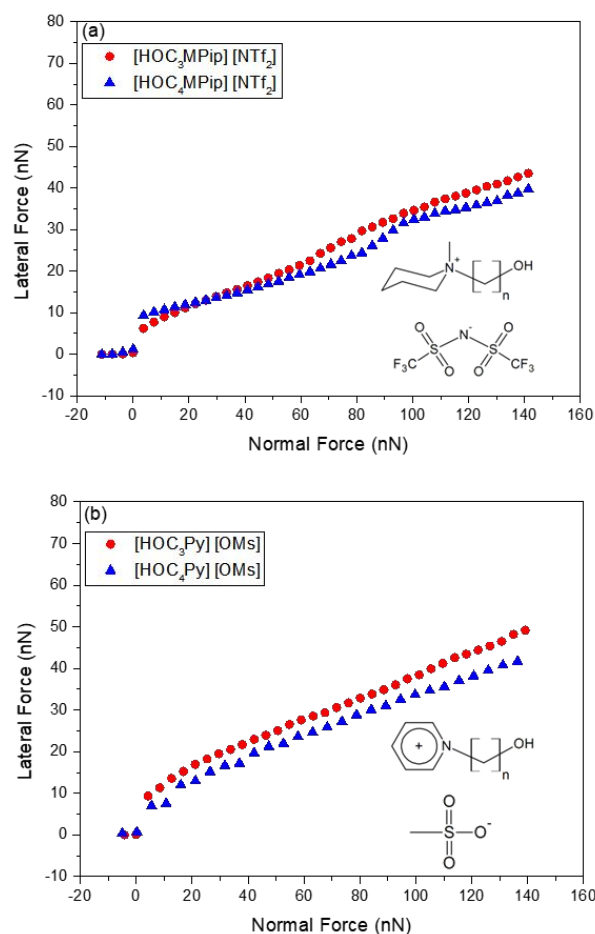


Figure 2. Lateral force vs normal force of (a) $[\text{HOC}_n\text{MPip}][\text{NTf}_2]$ and (b) $[\text{HOC}_n\text{Py}][\text{OMs}]$ at mica surfaces.

AFM nanotribology measurements have been used to measure boundary layer friction at the IL-mica interface for hydroxy-functionalized ILs, and the data are compared to a similar conventional IL. Hydroxy-functionalized ILs with weakly interacting anions form an unusual cation bilayer structure due to the formation of (c-c) H-bonds. These (c-c) H-bonds enhance ion order in the boundary layer which improves the lubricity. The strength of the cation bilayer structure can be tuned by altering the strength of (c-c) H-bonding via changing the hydroxyalkyl chain length, the cation charge polarizability and the coordination strength of the anions. [HOC₄Py][NTf₂] shows the most significant bilayer structuring at the mica interface because it has the strongest (c-c) H-bonding caused by the combination effects of weak coordinations between [HOC₄Py]⁺ cations and [NTf₂]⁻ anions, the strong charge delocalization of pyridium cation and a long hydroxyalkyl chain.

Acknowledgements

R.L. thanks the Deutsche Forschungsgemeinschaft (DFG) under grants LU 506/14-1 and LU 506/14-2 (286149019) for support of this work. T.N. thanks the “Deutscher Akademischer Austauschdienst” for supporting his research at the University of Western Australia, Perth. H.L. and R.A. acknowledge the Australian Research Council and the University of Western Australia for funding support.

Supporting Information. Calculation of contact stress, IR measurements

References

1. MacFarlane, D. R.; Tachikawa, N.; Forsyth, M.; Pringle, J. M.; Howlett, P. C.; Elliott, G. D.; Davis, J. H.; Watanabe, M.; Simon, P.; Angell, C. A., Energy Applications of Ionic Liquids. *Energy & Environ. Sci.* **2014**, 7, 232-250.
2. Baker, G. A.; Baker, S. N.; Pandey, S.; Bright, F. V., An Analytical View of Ionic Liquids. *Analyst* **2005**, 130, 800-808.
3. Earle, M. J.; Seddon, K. R., Ionic Liquids. Green Solvents for the Future. *Pure Appl. Chem.* **2000**, 72, 1391-1398.

4. Hayes, R.; Warr, G. G.; Atkin, R., Structure and Nanostructure in Ionic Liquids. *Chem. Rev.* **2015**, *115*, 6357–6426.
5. Endres, F.; Borisenko, N.; Al Salman, R.; Al Zoubi, M.; Prowald, A.; Carstens, T.; El Abedin, S. Z., Electrodeposition from Ionic Liquids: Interface Processes, Ion Effects, and Macroporous Structures. *Ionic Liquids Uncoiled* **2012**, 1–27.
6. Fumino, K.; Ludwig, R., Analyzing the Interaction Energies between Cation and Anion in Ionic Liquids: The Subtle Balance between Coulomb Forces and Hydrogen Bonding. *J. Mol. Liq.* **2014**, *192*, 94–102.
7. Niemann, T.; Zaitsau, D. H.; Strate, A.; Stange, P.; Ludwig, R., Controlling “Like–Likes–Like” Charge Attraction in Hydroxy-Functionalized Ionic Liquids by Polarizability of the Cations, Interaction Strength of the Anions and Varying Alkyl Chain Length. *Phys. Chem. Chem. Phys.* **2020**, *22*, 2763–2774.
8. Wilkes, J. S., A Short History of Ionic Liquids—from Molten Salts to Neoteric Solvents. *Green Chem.* **2002**, *4*, 73–80.
9. Forsyth, S. A.; Pringle, J. M.; MacFarlane, D. R., Ionic Liquids—an Overview. *Aust. J. Chem.* **2004**, *57*, 113–119.
10. McLean, B.; Li, H.; Stefanovic, R.; Wood, R. J.; Webber, G. B.; Ueno, K.; Watanabe, M.; Warr, G. G.; Page, A.; Atkin, R., Nanostructure of [Li (G4)] Tfsi and [Li (G4)] No 3 Solvate Ionic Liquids at Hg and Au (111) Electrode Interfaces as a Function of Potential. *Phys. Chem. Chem. Phys.* **2015**, *17*, 325–333.
11. Giernoth, R., Task-Specific Ionic Liquids. *Angew. Chem. Int. Ed.* **2010**, *49*, 2834–2839.
12. Elbourne, A.; Sweeney, J.; Webber, G. B.; Wanless, E. J.; Warr, G. G.; Rutland, M. W.; Atkin, R., Adsorbed and near-Surface Structure of Ionic Liquids Determines Nanoscale Friction. *Chem. Commun.* **2013**, *49*, 6797–6799.
13. Li, M.-G.; Chen, L.; Zhong, Y.-X.; Chen, Z.-B.; Yan, J.-W.; Mao, B.-W., The Electrochemical Interface of Ag(111) in 1-Ethyl-3-Methylimidazolium Bis(Trifluoromethylsulfonyl)Imide Ionic Liquid—a Combined in-Situ Scanning Probe Microscopy and Impedance Study. *Electrochim. Acta* **2016**, *197*, 282–289.
14. Page, A.; Elbourne, A.; Addicoat, M.; Warr, G.; Voitchovsky, K.; Atkin, R., 3-Dimensional Atomic Scale Structure of the Ionic Liquid–Graphite Interface Elucidated by Combined Am-Afm and Quantum Chemical Simulations. *Nanoscale* **2014**, *6*, 8100–8106.
15. Zhang, Q.; Wang, Q.; Zhang, S.; Lu, X.; Zhang, X., Electrodeposition in Ionic Liquids. *Chem. Phys. Chem.* **2016**, *17*, 335–51.
16. Zhang, Y.; Dyatkin, B.; Cummings, P. T., Molecular Investigation of Oxidized Graphene: Anatomy of the Double-Layer Structure and Ion Dynamics. *J. Phys. Chem. C* **2019**, *123*, 12583–12591.
17. Mao, X., et al., Self-Assembled Nanostructures in Ionic Liquids Facilitate Charge Storage at Electrified Interfaces. *Nat. Mat.* **2019**, 1–8.
18. Gu, Y.; Li, G., Ionic Liquids-Based Catalysis with Solids. State of the Art. *Adv. Synth. Catal.* **2009**, *351*, 817–847.
19. Li, H.; Endres, F.; Atkin, R., Effect of Alkyl Chain Length and Anion Species on the Interfacial Nanostructure of Ionic Liquids at the Au (111)–Ionic Liquid Interface as a Function of Potential. *Phys. Chem. Chem. Phys.* **2013**, *15*, 14624–14633.
20. Endres, F.; Borisenko, N.; El Abedin, S. Z.; Hayes, R.; Atkin, R., The Interface Ionic Liquid (S)/Electrode (S): In Situ Stm and Afm Measurements. *Faraday Discuss.* **2012**, *154*, 221–233.
21. Yokota, Y.; Harada, T.; Fukui, K.-i., Direct Observation of Layered Structures at Ionic Liquid/Solid Interfaces by Using Frequency-Modulation Atomic Force Microscopy. *Chem. Commun.* **2010**, *46*, 8627–8629.
22. Su, Y. Z.; Fu, Y. C.; Yan, J. W.; Chen, Z. B.; Mao, B. W., Double Layer of Au (100)/Ionic Liquid Interface and Its Stability in Imidazolium-Based Ionic Liquids. *Angew. Chem.* **2009**, *121*, 5250–5253.
23. Atkin, R.; Warr, G. G., Structure in Confined Room-Temperature Ionic Liquids. *J. Phys. Chem. C* **2007**, *111*, 5162–5168.

24. Elbourne, A.; McDonald, S.; Voïchovsky, K.; Endres, F.; Warr, G. G.; Atkin, R., Nanostructure of the Ionic Liquid–Graphite Stern Layer. *ACS Nano* **2015**, *9*, 7608-7620.
25. Li, H.; Wood, R. J.; Rutland, M. W.; Atkin, R., An Ionic Liquid Lubricant Enables Superlubricity to Be "Switched on" in Situ Using an Electrical Potential. *Chem. Commun.* **2014**, *50*, 4368-4370.
26. Sweeney, J.; Hausen, F.; Hayes, R.; Webber, G. B.; Endres, F.; Rutland, M. W.; Bennewitz, R.; Atkin, R., Control of Nanoscale Friction on Gold in an Ionic Liquid by a Potential-Dependent Ionic Lubricant Layer. *Phys. Rev. Lett.* **2012**, *109*, 155502.
27. Asencio, R. Á.; Cranston, E. D.; Atkin, R.; Rutland, M. W., Ionic Liquid Nanotribology: Stiction Suppression and Surface Induced Shear Thinning. *Langmuir* **2012**, *28*, 9967-9976.
28. Katsyuba, S. A.; Vener, M. V.; Zvereva, E. E.; Fei, Z.; Scopelliti, R.; Laurenczy, G.; Yan, N.; Paunescu, E.; Dyson, P. J., How Strong Is Hydrogen Bonding in Ionic Liquids? Combined X-Ray Crystallographic, Infrared/Raman Spectroscopic, and Density Functional Theory Study. *J. Phys. Chem. B* **2013**, *117*, 9094-9105.
29. Fakhraee, M.; Zandkarimi, B.; Salari, H.; Gholami, M. R., Hydroxyl-Functionalized 1-(2-Hydroxyethyl)-3-Methyl Imidazolium Ionic Liquids: Thermodynamic and Structural Properties Using Molecular Dynamics Simulations and Ab Initio Calculations. *J. Phys. Chem. B* **2014**, *118*, 14410-14428.
30. Panja, S. K.; Haddad, B.; Kiefer, J., Clusters of the Ionic Liquid 1-Hydroxyethyl-3-Methylimidazolium Picrate: From Theoretical Prediction in the Gas Phase to Experimental Evidence in the Solid State. *Chem. Phys. Chem.* **2018**, *19*, 3061-3068.
31. Niemann, T.; Zaitsau, D.; Strate, A.; Villinger, A.; Ludwig, R., Cationic Clustering Influences the Phase Behaviour of Ionic Liquids. *Sci. Rep.* **2018**, *8*, 1-7.
32. Strate, A.; Niemann, T.; Michalik, D.; Ludwig, R., When Like-Charged Ions Attract in Ionic Liquids: Controlling the Formation of Cationic Clusters by the Interaction Strength of the Counterions. *Angew. Chem., Int. Ed.* **2017**, *56*, 496-500.
33. Niemann, T.; Neumann, J.; Stange, P.; Gärtner, S.; Youngs, T. G. A.; Paschek, D.; Warr, G. G.; Atkin, R.; Ludwig, R., The Double-Faced Nature of Hydrogen Bonding in Hydroxy-Functionalized Ionic Liquids Shown by Neutron Diffraction and Molecular Dynamics Simulations. *Angew. Chem. Int. Ed.* **2019**, *58*, 12887-12892.
34. Elbourne, A.; Voïchovsky, K.; Warr, G. G.; Atkin, R., Ion Structure Controls Ionic Liquid near-Surface and Interfacial Nanostructure. *Chem. Sci.* **2015**, *6*, 527-536.
35. Hamilton, W. A.; Porcar, L.; Butler, P. D.; Warr, G. G., Local Membrane Ordering of Sponge Phases at a Solid--Solution Interface. *J. Chem. Phys.* **2002**, *116*, 8533-8546.
36. Niemann, T.; Li, H.; Warr, G. G.; Ludwig, R.; Atkin, R., Influence of Hydrogen Bonding between Ions of Like Charge on the Ionic Liquid Interfacial Structure at a Mica Surface. *J. Phys. Chem. Lett.* **2019**, *10*, 7368-7373.
37. Perkin, S., Ionic Liquids in Confined Geometries. *Phys. Chem. Chem. Phys.* **2012**, *14*, 5052-5062.
38. Perkin, S.; Crowhurst, L.; Niedermeyer, H.; Welton, T.; Smith, A. M.; Gosvami, N. N., Self-Assembly in the Electrical Double Layer of Ionic Liquids. *Chem. Commun.* **2011**, *47*, 6572-6574.
39. Zaitsau, D. H.; Neumann, J.; Niemann, T.; Strate, A.; Paschek, D.; Verevkin, S. P.; Ludwig, R., Isolating the Role of Hydrogen Bonding in Hydroxyl-Functionalized Ionic Liquids by Means of Vaporization Enthalpies, Infrared Spectroscopy and Molecular Dynamics Simulations. *Phys. Chem. Chem. Phys.* **2019**, *21*, 20308-20314.
40. Strate, A.; Neumann, J.; Niemann, T.; Stange, P.; Khudozhnikov, A. E.; Stepanov, A. G.; Paschek, D.; Kolokolov, D. I.; Ludwig, R., Counting Cations Involved in Cationic Clusters of Hydroxy-Functionalized Ionic Liquids by Means of Infrared and Solid-State Nmr Spectroscopy. *Phys. Chem. Chem. Phys.* **2020**, *22*, 6861-6867.
41. Ralston, J.; Larson, I.; Rutland, M. W.; Feiler, A. A.; Kleijn, M., Atomic Force Microscopy and Direct Surface Force Measurements. *Pure Appl. Chem.* **2005**, *77*, 2149-2170.

- 1
2
3 42. Alvarez-Asencio, R.; Thormann, E.; Rutland, M. W., Note: Determination of Torsional Spring
4 Constant of Atomic Force Microscopy Cantilevers: Combining Normal Spring Constant and Classical
5 Beam Theory (Vol 84, 096102, 2013). *Rev. Sci. Instrum.* **2014**, *85*.
6
7 43. Sweeney, J.; Webber, G. B.; Rutland, M. W.; Atkin, R., Effect of Ion Structure on Nanoscale
8 Friction in Protic Ionic Liquids. *Phys. Chem. Chem. Phys.* **2014**, *16*, 16651-16658.
9 44. Li, H.; Cooper, P. K.; Somers, A. E.; Rutland, M. W.; Howlett, P. C.; Forsyth, M.; Atkin, R., Ionic
10 Liquid Adsorption and Nanotribology at the Silica–Oil Interface: Hundred-Fold Dilution in Oil
11 Lubricates as Effectively as the Pure Ionic Liquid. *J. Phys. Chem. Lett.* **2014**, *5*, 4095-4099.
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60